NO.6 ENGINE DOES NOT RESTART [SKYACTIV-D 1.5]

id1103q4001100

6	ENGINE DOES NOT RESTART
DESCRIPTION	• The engine does not restart after the engine was stopped by the i-stop control.
POSSIBLE CAUSE	• Engine does not crank when engine restart is initiated — Engine starting system malfunction • i-stop (engine stop control) permit conditions are not met due to either of the following causes: — Driver operation — Environment (problem other than with vehicle) — i-stop related part malfunction **Note** • PIDs related to i-stop can be displayed in a list using the M-MDS data monitor diagnostic function. • If the engine is stopped by the i-stop control while the following data monitor items on the M-MDS are being viewed, the data monitor items may no longer be displayed. — BRAKE_SW — BRK_F_P_C — LON_ACCL_C — WSPD_SEN_LF — WSPD_SEN_RF — WSPD_SEN_LR — WSPD_SEN_RR

i-stop (engine restart control) condition

Detection module	Detection item	Determination conditions		PID	Inspection item(s)
		ATX	MTX		
PCM	Vehicle speed	1 km/h {0.6 mph} or more	4 km/h {2.5 mph} or more	WSPD_SEN_LF	• Front ABS wheel-speed sensor (See FRONT ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RF	
				WSPD_SEN_LR	• Rear ABS wheel-speed sensor (See REAR ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RR	
	Brake pedal	• Brake pedal is released → depressed while in P or N position	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
				BOO , BPA	• Brake switch (See BRAKE SWITCH INSPECTION.)
	Brake fluid pressure	Brake fluid pressure is 0.35 MPa {3.6 kgf/cm^2, 51 psi} or less while in D or M position	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
				BRK_F_P_C	• Brake fluid pressure sensor (See BRAKE FLUID PRESSURE SENSOR INSPECTION.)

Detection module	Detection item	Determination conditions		PID	Inspection item(s)
		ATX	MTX		
PCM	Accelerator pedal	Accelerator pedal is depressed while in D or M position	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
				APP	• APP sensor (See ACCELERATOR PEDAL POSITION (APP) SENSOR INSPECTION [SKYACTIV-D 1.5].)
	Clutch pedal	Not applicable	Depressed	CPP	• CPP switch (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-D 1.5].) • Clutch stroke sensor (See CLUTCH STROKE SENSOR INSPECTION [SKYACTIV-D 1.5].)
	Steering torque	4.0 N·m {41 kgf·cm, 35 in·lbf} or more while in D or M position	Not applicable	STR_TRQ_S_M	• EPS control module (See EPS CONTROL MODULE INSPECTION.)
	Steering angle	70° or more from center while in D or M position (After EPS control module learns center value)	Not applicable	STR_AB_EST	Not applicable
	Shift operation	Operated from P or N position to D, M, or R position	Not applicable	TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
	i-stop OFF switch	ON	←	I-Stop_OFF	• i-stop OFF switch (See i-stop OFF SWITCH INSPECTION [SKYACTIV-D 1.5].)
	Battery charge condition	65.7 % or less	←	BATT_SOC	• Current sensor (See CURRENT SENSOR INSPECTION [SKYACTIV-D 1.5].) • Battery (See BATTERY INSPECTION.)

Detection module	Detection item	Determination conditions		PID	Inspection item(s)
		ATX	MTX		
PCM	Power brake unit vacuum	−43 KPa {−0.44 kgf/cm^2, −6.2 psi} or more (value calculated by subtracting BARO from BBP)	←	BBP	• Power brake unit vacuum sensor (See POWER BRAKE UNIT VACUUM SENSOR INSPECTION [SKYACTIV-D 1.5].)
				BARO	• BARO sensor (See BAROMETRIC PRESSURE (BARO) SENSOR INSPECTION [SKYACTIV-D 1.5].)
	Driver not seated (except for European (L.H.D U.K.) specs.)	• All the following conditions are met: — Seat belt (driver's side): Not fastened — Door (driver's side): Open — P or N position	• All the following conditions are met: — Seat belt (driver's side): Not fastened — Door (driver's side): Open — Neutral	SEAT_B_D	• Driver-side buckle switch (See BUCKLE SWITCH INSPECTION.)
				DOOR_D	• Front door latch switch (RH) (See FRONT DOOR LATCH SWITCH INSPECTION.)
				TR	• Transaxle range sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
				CPP/PNP	• Neutral switch (See NEUTRAL SWITCH INSPECTION [SKYACTIV-D 1.5].)
	Engine stop period by i-stop	120 s or more	←	Not applicable	Not applicable
	Engine coolant temperature	23.5 °C {74.3 °F} or less	←	ECT	• ECT sensor (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR INSPECTION [SKYACTIV-D 1.5].)
	Fuel pressure	15 MPa {153 kgf/cm^2, 2176 psi} or less	←	FRP	• Fuel pressure sensor (See FUEL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].)
	System condition	An i-stop related part DTC is recorded	←	Not applicable	Not applicable
Start stop unit	+B (ROOM fuse) input	No	←	VPWR_B1	• Start stop unit (See START STOP UNIT INSPECTION.) • Battery (See BATTERY INSPECTION.)
	Remote transmitter authorization	When engine restart by remote transmitter authorization is requested	←	Not applicable	Not applicable
	System condition	i-stop, keyless entry system/advanced keyless entry system, push button start system and immobilizer system related part DTCs are not recorded	←	Not applicable	Not applicable

Detection module	Detection item	Determination conditions		PID	Inspection item(s)
		ATX	MTX		
TCM	Gear position	In 2nd gear fixed mode	Not applicable	Not applicable	Not applicable
	TFT	20 °C {68 °F} or less, or 117°C {243 °F} or more	Not applicable	TFT	• TFT sensor (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].)
	Vehicle speed	3 km/h {2 mph} or more	Not applicable	WSPD_SEN_LF	• Front ABS wheel-speed sensor (See FRONT ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RF	
				WSPD_SEN_LR	• Rear ABS wheel-speed sensor (See REAR ABS WHEEL-SPEED SENSOR INSPECTION.)
				WSPD_SEN_RR	
	System condition	An i-stop related part DTC is recorded	Not applicable	Not applicable	Not applicable
Climate control unit	A/C request (with full-auto A/C)	Yes	←	STOP_ST	• Climate control unit (See CLIMATE CONTROL UNIT INSPECTION [FULL-AUTO AIR CONDITIONER].) (See CLIMATE CONTROL UNIT INSPECTION [MANUAL AIR CONDITIONER].)
	A/C set temperature (with full-auto A/C)	Set to MIN	←	Not applicable	Not applicable
	MODE dial	DEFROSTER	←	Not applicable	Not applicable
	System condition	An i-stop related part DTC is recorded	←	Not applicable	Not applicable
DSC HU/CM	System condition	An i-stop related part DTC is recorded	Not applicable	Not applicable	Not applicable

Diagnostic Procedure

Step	Inspection	Results	Action
1	**VERIFY ENGINE CRANKING OCCURRENCE** • Start the engine. • Does the engine start?	Yes	Go to the next step.
		No	Perform symptom troubleshooting according to NO.9 WILL NOT CRANK and NO.12 CRANKS NORMALLY BUT WILL NOT START. (See NO.9 WILL NOT CRANK [SKYACTIV-D 1.5].) (See NO.12 CRANKS NORMALLY BUT WILL NOT START [SKYACTIV-D 1.5].)

Step	Inspection	Results	Action
2	**VERIFY DTCs** • Using the M-MDS, display the DTCs for the PCM, TCM, DSC HU/CM, climate control unit, and start stop unit. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].) (See DTC INSPECTION [DSC HU/CM].) (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) (See START STOP UNIT INSPECTION.) • Is a DTC displayed?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].) (See DTC TABLE [DSC HU/CM].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].) (See DTC TABLE [START STOP UNIT].)
		No	Go to the next step.
3	**VERIFY i-STOP (ENGINE RESTART CONTROL) PERMISSION CONDITION ITEM** **Warning** **• To prevent an accident, work with two people when the vehicle is being driven (one person drives vehicle, other person operates M-MDS).** **• Observe legal speed limits when driving the vehicle and perform the procedure in a location where safety can be assured.** • Switch the ignition ON (engine off). • Display the PID items related to i-stop using the M-MDS. (See PID/DATA MONITOR INSPECTION (i-stop) [SKYACTIV-D 1.5].) • Drive the vehicle and perform engine stopping by i-stop. • Verify if the i-stop (engine restart control) permit conditions of the items which can be verified by the PID are met. • Are the permit conditions of all the items met?	Yes	Go to the next step.
		No	For the applicable items, inspect the parts under Inspection Item(s).
4	**VERIFY VEHICLE CONDITIONS RELATED TO i-STOP (ENGINE RESTART CONTROL) PERMISSION CONDITIONS** • Drive the vehicle and perform engine stopping by i-stop. • Verify that the engine restarts when any of the following conditions is met: — Gear position: 2nd gear fixed mode — A/C set temperature: MAX or MIN — Windshield defroster switch: ON • Does the engine restart?	Yes	Verify the i-stop (engine restart control) permission conditions, then go to Step 11.
		No	Go to the next step.
5	**INSPECT AMBIENT TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the ambient temperature sensor. (See AMBIENT TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) (See AMBIENT TEMPERATURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) • Is the ambient temperature sensor normal?	Yes	Go to the next step.
		No	Replace the ambient temperature sensor, then go to Step 11. (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)

Step	Inspection	Results	Action
6	**INSPECT EVAPORATOR TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the evaporator temperature sensor. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) • Is the evaporator temperature sensor normal?	Yes	Vehicles with full-auto A/C • Go to the next step. Vehicles with manual A/C • Go to Step 9.
		No	Replace the evaporator sensor, then go to Step 11. (See EVAPORATOR TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See EVAPORATOR TEMPERATURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)
7	**INSPECT CABIN TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the cabin temperature sensor. (See CABIN TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the cabin temperature sensor normal?	Yes	Go to the next step.
		No	Replace the cabin temperature sensor, then go to Step 11. (See CABIN TEMPERATURE SENSOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
8	**DETERMINE IF MALFUNCTION CAUSED BY BAD AIR MIX ACTUATOR** • Inspect the air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the air mix actuator normal?	Yes	Inspect the air mix door and linkage for sticking. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • If the steering switch is normal: — Go to the next step. • If there is any malfunction: — Repair or replace the malfunctioning location, then go to Step 11.
		No	Replace the air mix actuator, then go to Step 11. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
9	**DETERMINE IF MALFUNCTION CAUSED BY BAD AIRFLOW MODE ACTUATOR** • Inspect the airflow mode actuator. (See AIRFLOW MODE ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the airflow mode actuator normal?	Yes	Go to the next step.
		No	Replace the airflow mode actuator, then go to Step 11. (See AIRFLOW MODE ACTUATOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
10	**INSPECT EPS CONTROL MODULE FOR MALFUNCTION** • Inspect the EPS control module. (See EPS CONTROL MODULE INSPECTION.) • Is the EPS control module normal?	Yes	Implement the following procedure, then go to the next step. 1. Switch the ignition to OFF, and maintain this condition for approx. 3 s. 2. Set the vehicle wheels in the straight-ahead position. 3. Switch the ignition ON (engine off or on).
		No	Replace the EPS control module, then go to the next step. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
11	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Has the malfunction symptom been eliminated?	Yes	Complete the symptom troubleshooting. Explain repair contents to customer.)
		No	Repeat the inspection from Step 1.